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# A review on multimodal medical image fusion towards future research

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**Abstract**
Image fusion is a technique used to merge two or more source images into a single image that incorporates more details than the originals and still offering an accurate depiction about the captured information. Resultant fused images are more accurate and provide comprehensive information for both the human and machine vision perception for further processing of the image. Image fusion provides better performance in the areas like pattern recognition, image processing, computer vision, machine learning and artificial intelligence. In the recent years image fusion has moved out of the laboratories and used in the real time applications. This paper provides the insight of various techniques for image fusion like primitive fusion (Simple averaging, Maxima and Minima, etc.), Discrete Wavelet Transform (DWT) based fusion, Principal Component Analysis (PCA) based fusion, Curvelet transform based fusion etc. On-going through various literatures, it is found that image fusion in spatial domain provides high resolution images, although the fusion algorithms are dependent on the nature of image and also depends on the application for which the image is to be fused. Hence, spectral domain fusion and hybrid fusion techniques are introduced and it is proven to be better than the spatial domain fusion. Comparison of all the techniques along with recent approaches are done to find the best approach towards future research to provide new direction to the researchers in medical sector.

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## 1 Introduction

Medical images are the reflective information from the human body or organs; physiological organs are the unit of group of cells. Since, biomedical diagnostics and analysis are much needed in medical field; medical images and image related works are growing in day-to-day research. Since 1990s, a large number of literatures in journals, magazines, and conferences have been published and helps the research for further development. Computer aided systems are rapidly growing to achieve low cost and best image techniques [39], which leads to various imaging modalities and various algorithms for medical image analysis and diagnosis. Each imaging modalities are the primary source of medical analysis and diagnosis, also each imaging source having particular information pertaining to its frequency range and advancement in the imaging technology. To be particular, it is not possible to capture all the information about human body, organs and cells with the help of single imaging modality. Each modality has its own set of advantages: X-rays are ideal for bone scanning, CTs provide information on internal organs and soft tissues, and MRIs allow for a closer examination of organs and soft tissues, PET scans offer information on blood flow, oxygen intake, and organ metabolism. As MRI scanning takes more time, it is not suggested by physicians as a first line of defence for any diagnosis. Initially, physicians recommend CT scans because they are speedier and can aid in first-level diagnosis. Only in cases when a CT scan proved insufficient information, an MRI is recommended. Each modality adds value to the existing data, hence combining the two or more images from different modalities for better diagnosis is recommended. This data fusion can be accomplished by either combining image features or combining the image itself.

Multimodal Image fusion is a process of combining or converging multiple source image information from various modalities to obtain fused image as single image for improved clinical diagnosis. An example, two matrices are considered for spatial domain fusion as CT and MRI images as shown in Fig. 1. The maximum pixel value in an image is assumed to be a significant information. It is needed to fuse both the matrix together by considering the maximum pixel fusion rule.

Algorithm:

Step 1: Read the source images as matrix.

Step 2: Fix the fusion rule (Maximum, Minimum and Average).

Step 3: Find the greatest of two pixels belongs to the same index.

Step 4: Constitute new matrix by considering the maximum pixel.

Step 5: Display the new matrix as fused image.

The most required algorithm needs to be formulated to have various source image information together, which is achieved by multimodality image fusion. Multimodal performs better when compared to Multiview, multifocus. Multimodal image fusion incorporates images from various modality to provide more accurate knowledge about the image content, allowing for clearer analysis and more reliable performance. Additionally, integrating multi-modal images has many benefits, including complementing data for better classification, improving geometric corrections, and enhancing features for study, among other fusion approaches [89]. The process of integrating relevant information from various source images into a single image

| | | |
|---|---|---|
| 5 | 4 | 7 |
| 1 | 3 | 2 |
| 7 | 6 | 5 |

**(A)**

| | | |
|---|---|---|
| 4 | 6 | 1 |
| 3 | 4 | 7 |
| 6 | 5 | 2 |

**(B)**

| | | |
|---|---|---|
| 5 | 6 | 1 |
| 3 | 4 | 7 |
| 7 | 6 | 5 |

**(C)**

**Fig. 1** Image of size 3 × 3 **A** CT image **B** MRI image **C** Fused image

with more descriptive and accurate than the input source images is called as image fusion. Image fusion aims to generate images that are more suitable and understandable for both humans and machines. Image fusion is not only used to fuse information and also to reduce redundant amount of data from multiple sources. Image fusion is classified into three types based on domain; spatial domain, transform domain and hybrid domain image fusion. Methods like pixel averaging, maxima and minima, Brovey, PCA, High Pass Filtering (HPF), and Intensity -Hue Saturation (IHS) comes under spatial domain. Figure 2 depicts the various approaches reviewed in the study. To the extend, this survey paper includes a large number of articles published in various journals and conferences since 2014. Multispectral MR pictures aid in obtaining physiological information about a patient with a brain infarction [50]. Since MR images contain more detailed information in the spectrum domain, they are usually studied in conjunction with other modality images. In 2019, CT and MRI image fusion is used to analyse the kidney lesion in [9]. Wavelet transform, complex wavelet transform, Convolutional Neural Network (CNN), Non-subsampled Contourlet Transform (NSCT), and other image fusion techniques are used to develop various fusion approaches.

Fusion approaches based on Discrete Cosine Transform (DCT), Discrete Wavelet Transform (DWT), Curvelet transform, Dual Tree Complex Wavelet Transform (DTCWT), Stationary Wavelet Transform (SWT), etc. comes under transform domain. Both the spatial and transform domains are combined and proposed many methods like ST and PCA [64], DTCWT with PCA [2], cascaded PCA and shift invariant wavelet transform [13] are the advancement in fusion. High resolution multispectral image had been obtained by fusing two modality images. To improve the image quality, the same modality or multiple modality images were fused; multiple modalities such as Photon Emission Computed Tomography (SPECT), Computer Tomography (CT), Magnetic Resonance Angiogram (MRA), Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), infrared, visible, and thermal images are used.

From 2000 to 2020, medical image fusion technology has been developed significantly, as shown in Fig. 3 the number of published papers of medical image fusion has remarkably boomed in the recent years (medical image fusion publications were counted).

### 1.1 Image registration

Image registration is the first step in bringing the source input images into spatial alignment with each other [24]. Registration is the process of aligning one source image with another source image by considering image functions such as intensities, colours, and other characteristics in an overlapped frame area. It's a crucial step in multimodal medical image fusion

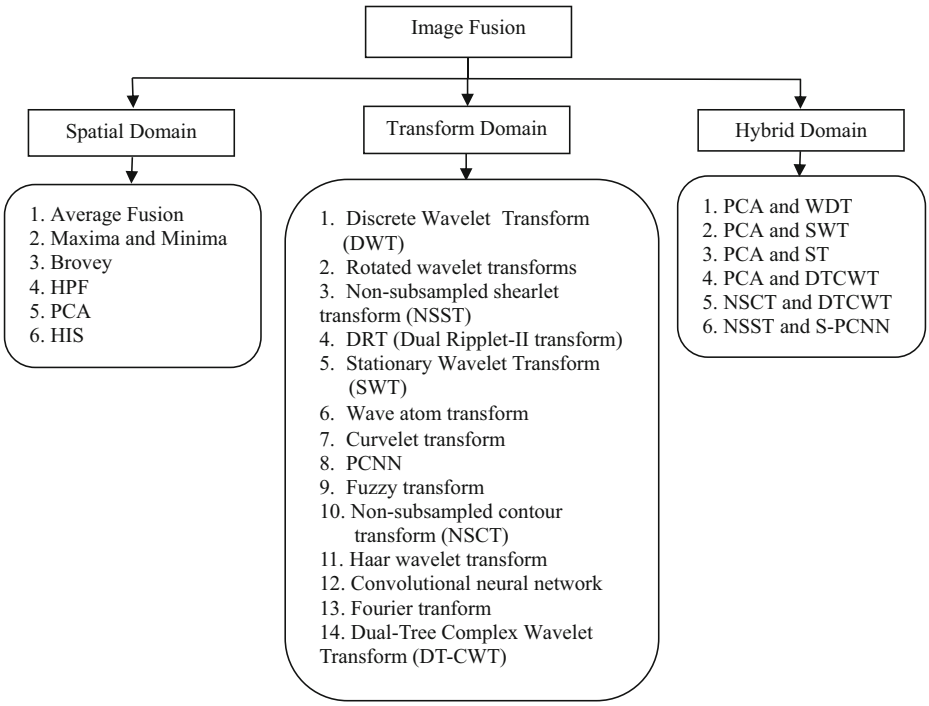


**Fig. 2** Various approaches of medical image fusion

because it improves the ability to combine information from different modalities' images [18]. Medical images are taken at various times, which might cause geometric deformations. To fuse two or more images together, image registration is used to reduce linear and non-linear discrepancies between the source images. Medical image registration is the process of aligning

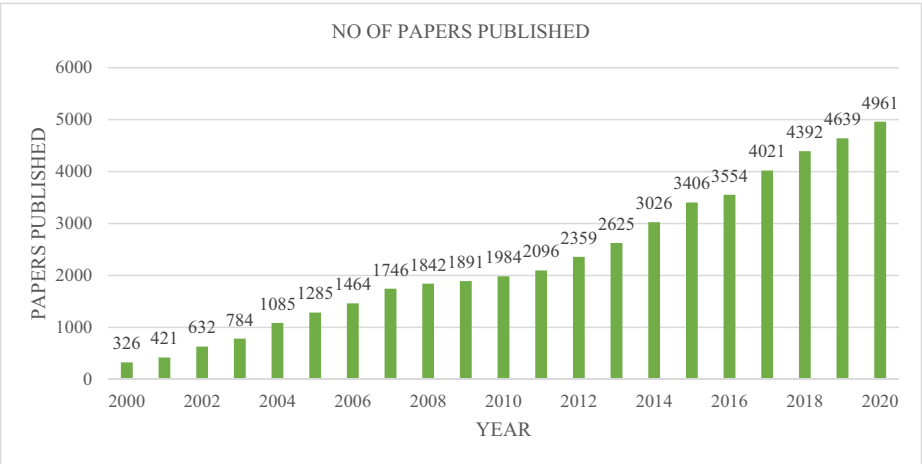


**Fig. 3** Papers published in the recent years

one image data with the equivalent points in another image data. Area-based and Feature-based approaches are the two basic types of picture registration techniques.

Image registration works usually in four steps

i) Feature detection: In source image and reference image salient features or distinct objects like line, edges are identified.
ii) Feature matching: Mapping of features between the reference image and source image for transformation.
iii) Transform model estimation: Align the source image with respect to reference image based on the mapped feature in feature mapping step.
iv) Image resampling and transformation: Source image is transformed by means of mapping function.

The general block diagram of image fusion is shown in Fig. 4.

### 1.2 Feature detection

The typical features of the registered images are extracted and more than one feature maps are produced for each of the input images at this stage. Feature extractions are done either in spatial domain or in spectral domain. Deep neural network is used to extract the depth features by current researches.

### 1.3 Fusion rules

The fusion rules are depending on the fusion domain. In this approach, the maximum pixel value or minimum pixel value or average value to be considered in pixel level fusion. In spectral domain, features are extracted and rules to be framed to consider best feature to retain in fused domain.

## 2 Medical imaging modalities

In fusion process, different imaging modalities are used and they are described in brief in this article. In this the fusion can be done by combing the same modality together or different modality images. The different modalities are MRI, PET, CT, SPECT, MRA. Figure 5 depicts the sample images of different modalities.

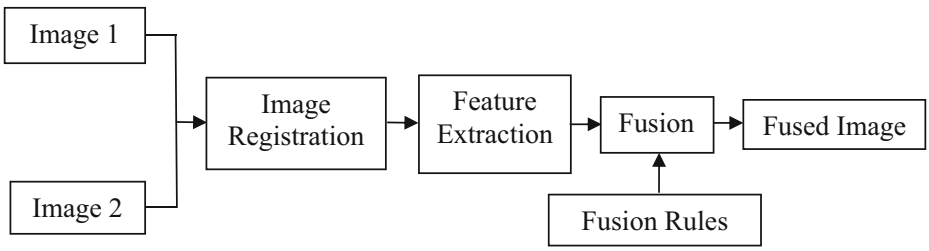


**Fig. 4** General block diagram of image fusion

### 2.1 Magnetic resonance imaging

MRI is important because it gives clearer and provides important information like brain, spin, joints, etc. Since it gives accurate results, MR images is used to fuse with other modality images. In [7, 8, 19, 29, 32, 56, 57, 59, 65, 66], MR images are used with different modality

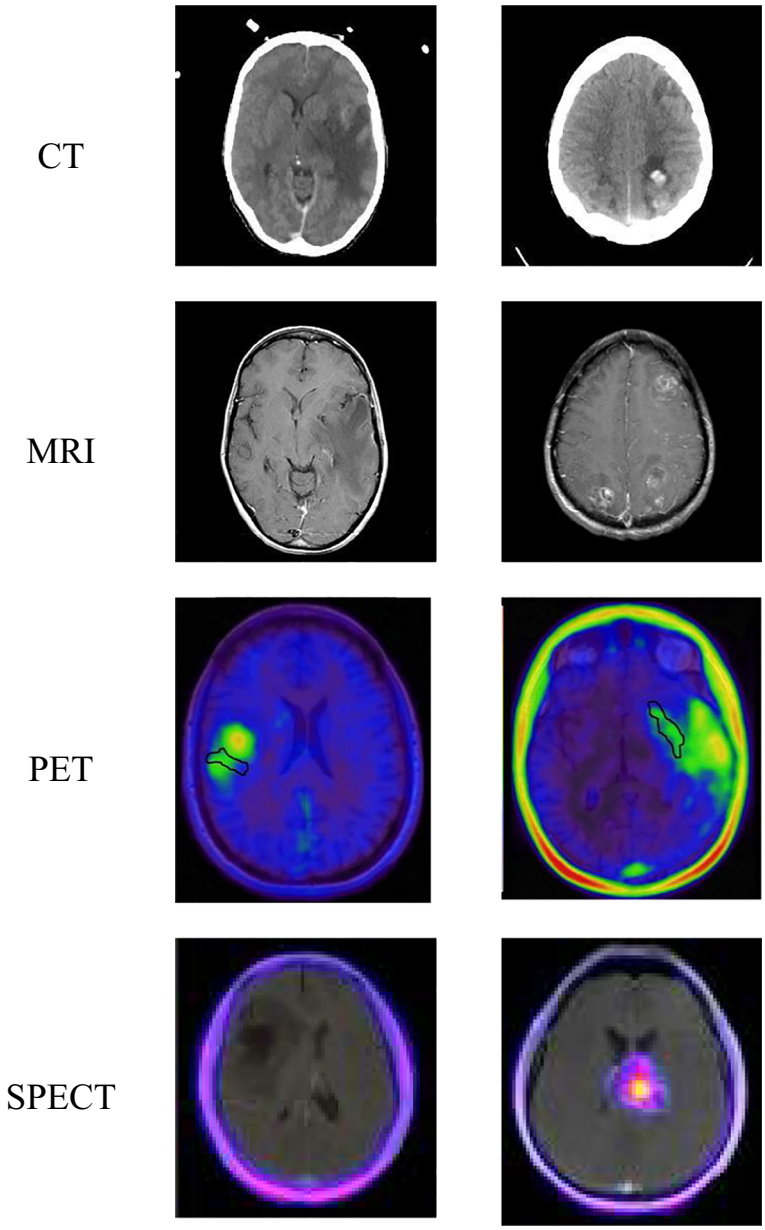


**Fig. 5** Images of different modalities

for various applications like brain tumour detection [54], assessment of kidney lesion [89], diagnosis for stomach cancer [72], vascular blood flow diagnosis.

### 2.2 Positron emission tomography

In addition to anatomical information, this imaging provides functional information and metabolisms of specific tissues. The images are visually clear and does not provide artifacts. Many researchers have used PET images for different applications such as cancer staging, recurrent tumor detection [16], location of lesion [23], etc.

### 2.3 Computer tomography

The tomographic images of specific areas of the body can be obtained and the cross-sectional images are used in various medical applications [38, 63, 67], such as diagnosis of lung cancer.

### 2.4 Single photon emission computed tomography

The SPECT image is a functional image that shows the metabolism of human tissue, organs, and blood flow through arteries and veins, but low image resolution and poor positioning capacity [48]. This imaging is used for a number of reasons, including identifying lesions [34] and detecting vertebral bone metastasis in tinnitus patients.

## 3 Multimodal medical image fusion

Several image fusion techniques are developed since 1985 and recent successful techniques are discussed in this review paper.

### 3.1 Spatial domain

The IHS transform [19, 84] transforms Red, Green, and Blue (RGB) colour image into independent IHS components. In IHS transform the brightness of a spectrum is defined by the intensity, the hue by the spectral wavelength, and the saturation by the spectrum's purity. The PET and MRI images are used as input images, to integrate retina inspired and IHS fusion techniques [19]. The histogram of MRI image was compared with the histogram of PET-intensity portion, and it was found that the colour information was least distorted and the spatial information was as clear as the original MRI image, indicating that the integration of colour and spatial features provides good visual quality. The general block diagram for the spatial domain transform is shown in Fig. 6. The spatial features are obtained by using PCA which decreases the dimension of the data and also reduce the redundancy in both the input source images [13, 84]. A collection of correlated variables is transformed into a set of uncorrelated variables using this method. The PCA and pyramid fusion techniques are combined, in which both the pixel level and region-based fusions were used [56].

The spatial domain image fusion technique process is described in flowchart for easy understanding in Fig. 7. The structure tensor proved to be easy to detect the lesion region by fusing MRI and PET images [23]. The average filter and the structure tensor were used to decompose the images into a two-scale image representation. Then, for coarse and informative

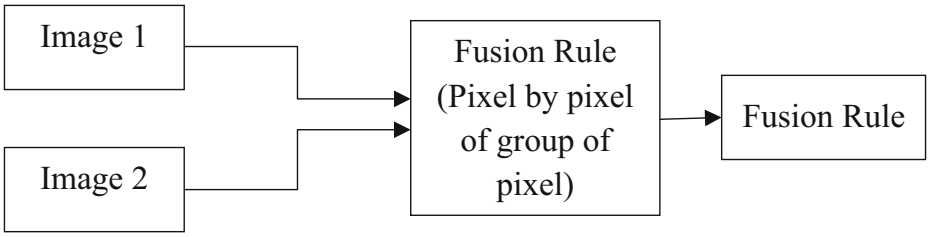


**Fig. 6** General block diagram of spatial domain fusion

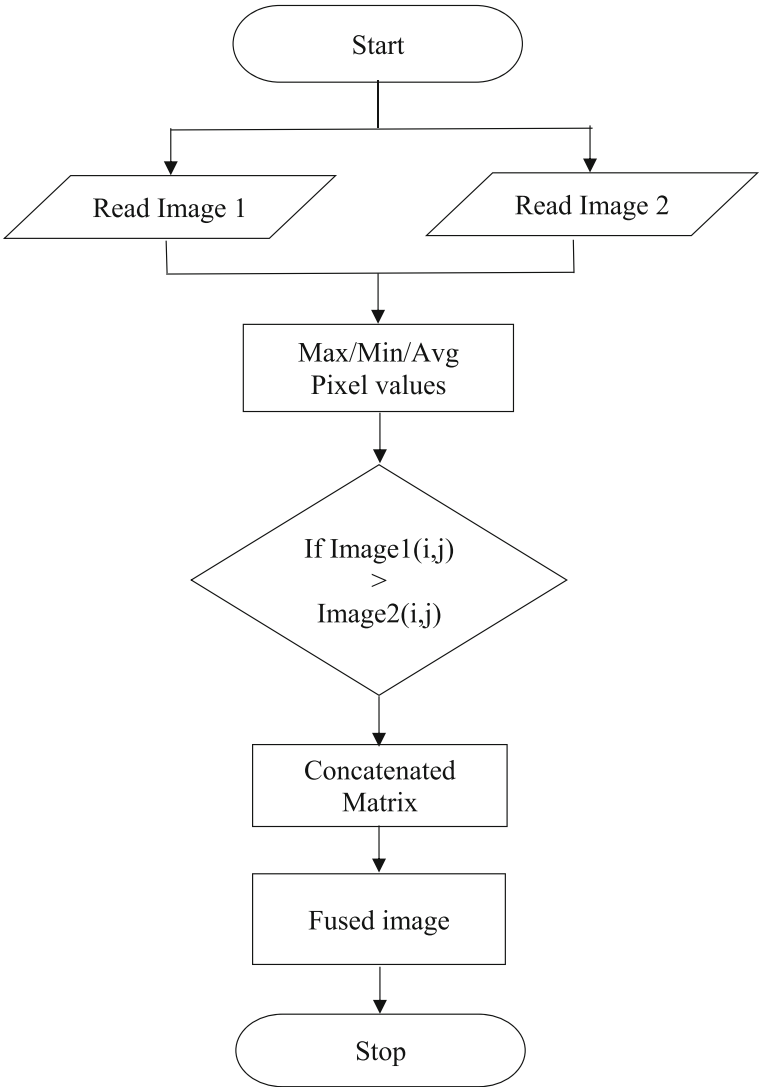


**Fig. 7** Flowchart of spatial domain fusion

layers, the average and local energy were considered as fusion laws. The coarse stands of light matter in the human brain are precisely differentiated in the process of MRI-SPECT fusion, as well as the texture and edge details retain from the source image. Table 1 gives the remarks of various spatial domain techniques.

As a result, spatial domain was used due to easy computation, direct manipulation from the source images. But the drawback was only less quality was achieved.

### 3.2 Transform domain

A comparative analysis is done on various transform domain fusion for medical applications which act as a guide to this section [6]. To minimise computational complexity, DCT-based fusion is used, in which the images are first decomposed into a series of waveforms and then fused [51]. If the block size is less than $8 \times 8$ or equal to the image size, the main fused image would be of low quality. SWT along with the Spatial Frequency Measurements (SFM) was used, which provided good result at the 2nd level of decomposition but the process consumes more time [77]. Discrete Fractional Wavelet Transform (DFRWT) was established to fuse multimodal medical images [83]. The schematic diagram for the transform domain is shown in Fig. 4. In which the image2 is registered to adjust the orientation in line with the image1 and image2 are registered to each other. In the Fig. 8 LF represents the Low Frequency and the HF indicates the High Frequency.

Shift-Invariant Dual-Tree Complex Shearlet Transform (SIDCST) was used to fuse infrared and visible spectral images by integrating DTCWT and Shearlet Filter (SF) together [86]. SF and Dual-channel adaptive pulse coupled neural network based fusion rules are applied. A fusion framework was proposed by combining the Multi Scale Transform (MST) and SR to overcome the inherent defects of both MST and SR based fusion methods simultaneously. The Non-Subsampled Shearlet Transform (NSST), which is a multidimensional wavelet extension, combines the Non-Subsampled Laplacian Pyramid (NSLP) filter with the shearlet transform to provide multiscale decomposition [32, 38]. Multiscale decomposition is the direction of localization that employs a modified shearlet filter to map from pseudo polarisation to Cartesian coordinate system without requiring a subsequent sampling operation via Fourier Inverse Transform, and it has the shift-invariance property to retain the salient image features and detailed information of multiple MRI and CT source images to produce an enhanced

**Table 1** Remarks on spatial domain techniques

| Fusion techniques | Observation from the literature |
|---|---|
| Average [62, 69] | Reduces computational complexity but it has less guarantee for the resultant image to have good visual clarity (quality) |
| Maxima and Minima[62] | Focused image but produce blurring effect |
| Brovey method [69] | It's a very simple, efficient computation that takes less time to process. It generates RGB images with a higher contrast ratio. However, spectral degradation occurs. |
| IHS [19, 53, 69, 84] | Separate channels, high sharpening ability but only three bands (RGB) are involved and also provide colour distortion |
| HPF [53] | Achieves less quality index and produce spectral degradation |
| PCA [13, 56] | Can reduce the dimension without much loss in the information but produce spectral degradation and colour distortion. |

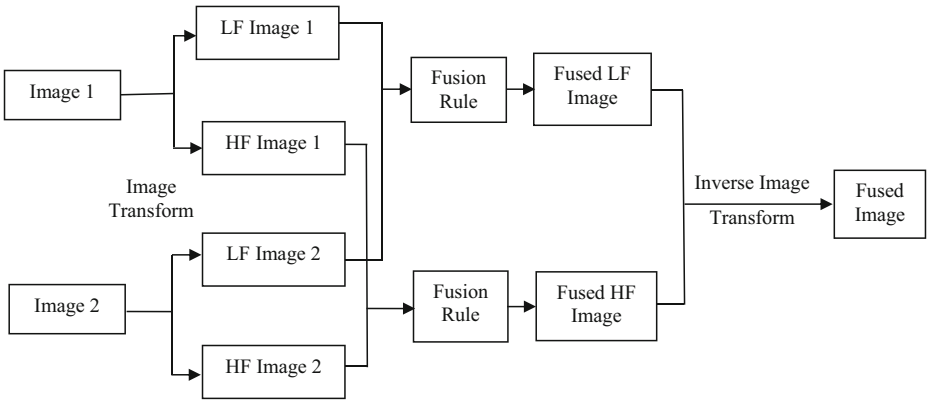


**Fig. 8** Schematic diagram of transform domain fusion

single fused image. Structure tensor and NSST were proposed for fusion [46]. The general flowchart for transform domain image fusion is shown in Fig. 9.

In the NSST domain, a Convolution Neural Network (CNN) and a Dual Channel Spiking Cortical Model (DCSCM) are utilised together to fuse the images together. The method was implemented using Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) images, which provide more data, enhanced contrast while maintaining the salient features, and synthesises soft tissue and bone structure information to the greatest extent possible [32]. Adaptive PCNN in NSST domain was used [27, 37] to fuse anatomical images like CT-MRI and SPECT-MRI to overcome the drawbacks of NSST based fusion in [74]. Orientation Information Motivated Pulse Coupled Neural Networks (PCNN) and modified decision map were used in which the final fused image can be obtained by selecting the pixels in the focus areas [22]. In the fusion process, CT, MRI, and PET images were used, and the fusion was based on NSST, S-PCNN, and ICM in HSV colour space [38]. The features of CT and MRI images are first decomposed into sub-bands coefficient using NSST, and then the fused sub-band coefficients of CT and MRI images are obtained using fusion rules based on the OFG of ICM1 and S-PCNN1. Second, the PET images are converted to HSV colour space, and then NSST decomposes the V element. Third, fusion rules based on the OFG of ICM2 and S-PCNN2 fuse the fused coefficients of CT-MRI and the decomposed coefficients of the V component. Finally, the inverse HSV and NSST transformations were used to create a fused multimodal medical image respectively and it was determined that it contains more detail than the original image.

In SAR and PAN, NSCT-RLMAP-MESMM fusion was introduced, in which the edges and contours are well captured contours and the texture information improves the image resolution by shift-invariance [5]. To fuse images from modalities like CT, MRI, PET, MRA, and SPECT, the quick curvelet transform and the Genetic Algorithm (GA) were used [8]. To achieve a fast computational operation, GA was used to optimise the function for fusion. It was shown that the fast curvelet transform is free from the directions for decomposition and construction. The MRI and CT images are used as inputs for fusion process [55].

In fusion, Discrete Wavelet Transform (DWT) along with the Binary Crow Search Algorithm (BCSA) were utilized to optimise the weight. The NSCT fusion technique was used to fuse the MRIT1 and MRIT2 images and the MRIT2 and CT images. The algorithm combines the advantage of NSCT and the image visual saliency [42]. The input modality for fusion was

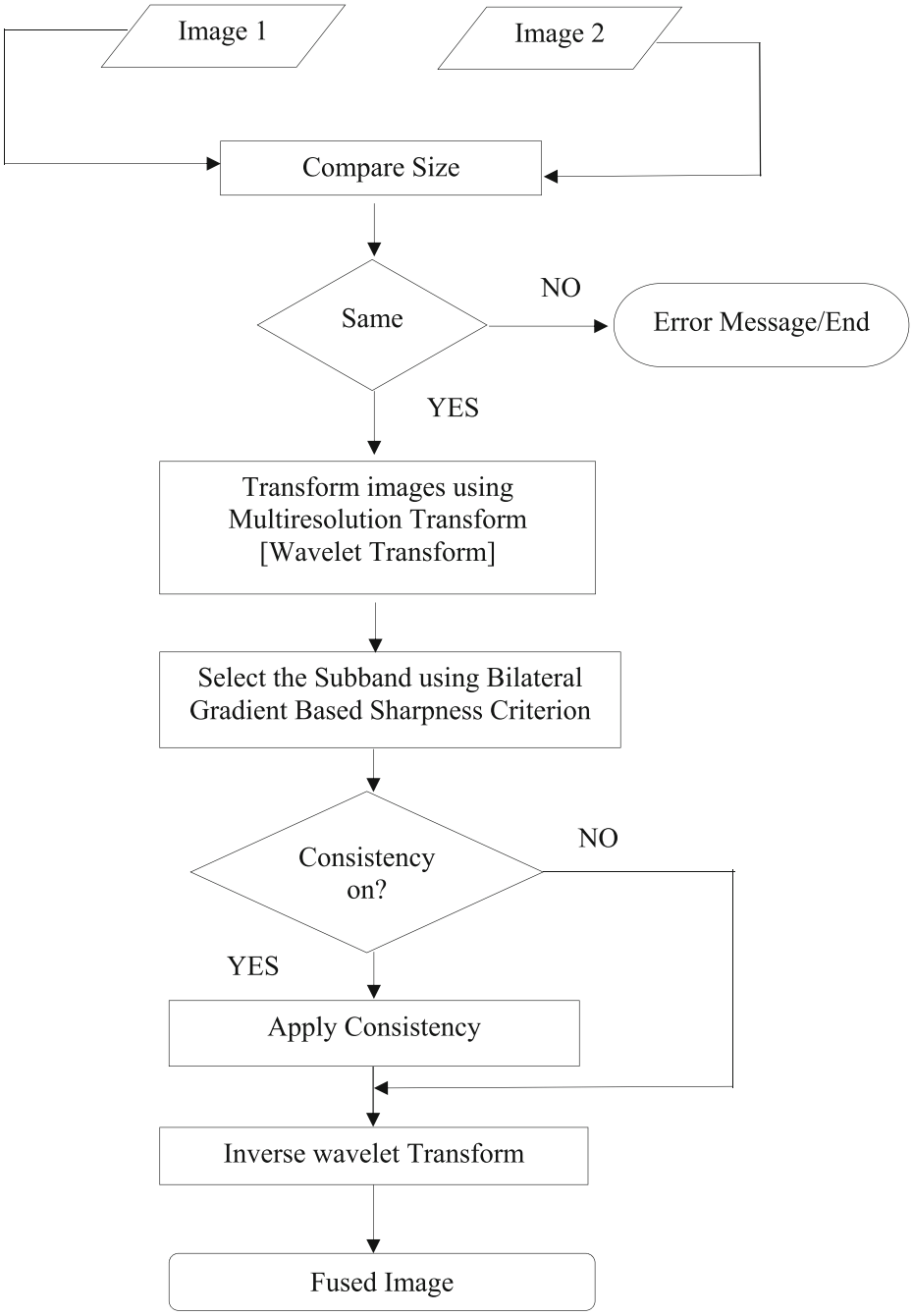


**Fig. 9** Flowchart of transform domain fusion

CT, MRI, T1-weighted MR (MR-T1), MR-Generalized Anxiety Disorder (GAD), and PET photos. In which the NSCT technique was used, it was found that the NSCT-based approach was superior to pixel averaging and DWT-based approaches in terms of providing more detail

and higher contrast than the input medical images [28]. The computing time is high, but obtains distinct and more natural fused image while using NSCT technique for multimodal images like (1) CT and MR, (2) MR-T1 and MR-T2 [14, 15]. NSCT with type-2 fuzzy logic technique was used to fuse MRI and CT images [85].

Rotated Wavelet Transform (RWT) was used for fusion approach which is used to fetch the edge-related features of MRI and CT [17]. A PCNN in NSCT domain was used to fuse CT and PET images [71]. A fusion method is proposed to combine SPECT and CT images, by combining Shuffled Frog Leaping Algorithm (SFLA) and the Pulse Coupled Neural Network (PCNN) to improve the quality of the resultant image [34].

Gradient Transfer Fusion (GTF) is a technique for fusing infrared and visible modalities because it retains thermal information through radiation with the help of infrared and retaining structural information with the help of visible spectral. The results appeared as high-resolution infrared images with clearly highlighted targets, making it useful for fusion-based target detection and recognition systems [47]. MCFO optimize wavelet based fusion algorithm developed using Haar wavelet and histogram matching technique to fuse various modality images like (1) MRI and PET (2) CT and PET (3) MRI and CT and gave the peerless gain parameters that achieve the best quality of fused picture [24, 25]. An algorithm to fuse the MRI and CT images was developed namely latent low rank representation and gradient transfer based fusion [49] employed by NSCT. Latent low rank representation is a tool used to separate detailed information and the saliency information.

MRI and CT images were registered using Scale-Invariant Feature Transform algorithm [3] and wavelet transform and curvelet transforms are used to fuse source images. Additionally, an interpolation was added to enhance the resolution of the fused image. Deep Belief Network (DBN) and fuzzy fusion rules was utilized to fuse images. At first DBN was applied to both CT and MRI-T1 images, in which the image is informative and non-informative blocks were distinguished using DBN [40]. To obtain a fused image, fuzzy based fusion rules were used, which preserve the informative blocks from the input images and also removes the redundant blocks.

Fusion based on sparse representation and cartoon texture decomposition was deployed to decompose the source images into texture and cartoon components [87]. An energy-based fusion rule was used, to retain the structure details of cartoon components, A Sparse Representation (SR) based fusion method was used for texture components. A dictionary learning method was applied to the source images in the SR-based fusion method, and image features were grouped into categories using SKR attributes and the LDP clustering method. First images are grouped as sub-dictionary for sparse representation, these sub-dictionaries with various traits are merged together as a comprehensive dictionary. The findings were found to be stronger and took less time when fused different multi-modal medical images (MRI and PET, MRI and SPECT), infrared-visible images, and multi-focus (colour and grey level images) images.

The wavelet transform has been found to have good time frequency localization and is used to execute multi-scale and multi-resolution operations [6]. The DWT has poor directionality, shift sensitivity, and the lack of phase information, have prompted the creation of new wavelet transforms [70]. The Dual Tree Complex Wavelet Transform (DT-CWT) is one among these prompted model (DT-CWT). The DT-CWT enhances, the image directionality by expanding the DWT using complex filter coefficient. The complex-valued filtering is used to decompose the image into real and imaginary parts representing the phase and magnitude components. As a consequence, DT-CWT is a DWT with significant additional properties including shift

invariance and selective directionality in two or more dimensions. Only for d-dimensional signals does it achieve these benefits with a redundancy factor of 2d. It is based on a Hilbert transform and a complex filter bank [52]. Intensity based image registration was used to register the images for fusion process. There are two datasets: MRI and CT images of a brain tumour, as well as MRI-T1 weighted (non-contrast) and MRI-T2 weighted images. The DT-CWT fusion technique uses the Modified Central Force Optimization (MCFO) to optimise the fusion gains [24]. The MCFO had demonstrated a high degree of accuracy and fast processing times while fusing medical images.

For disintegration and restoration of the source images, a lifting-based Biorthogonal Wavelet Transform (BWT) was used, as well as two fusion rules: Average fusion rule and Absolute Maximum fusion rule are adopted [60]. Edge information are obtained, in the fused image. FFST-SR-PCNN, a medical image fusion algorithm based on the Fast Finite Shearlet Transform (FFST) and sparse representation (SR) [73], was implemented to extract finer contour information from the edge of images, highlight edge features, and obtain more abundant information. The FFST-SR-PCNN decomposes the recorded source images into low and high frequency coefficients using FFST, then fuses the low frequency coefficients using the SR fusion algorithm and the high frequency coefficients using the simplified PCNN model's algorithm. Finally, the inverse FFST was used to recreate the fused low- and high-frequency coefficients in order to obtain the fused image.

The Daubechies Complex Wavelet Transform (DCxWT) was implemented, in which the images are decomposed to obtain coefficients that retain multiscale edge information as well as phase information. The maximum selection rule for fusion was used because higher value wavelet coefficients hold valuable information about the source images such as edges, corners, and so on. To fusion phase information, various types of images were used, including MRI and CT, T1 weighted MR image, and MRA. MRA images can detect anomalies, but due to their poor spatial resolution, they are unable to provide information about tissues. Hermite Transform (HT) was introduced to fuse SPECT and CT images [12]. HT is a tool which can decompose an image into set of coefficients defined in the space of Hermite polynomials. Review on various transform domain techniques is listed in Table 2.

As observed, the transform domain was used due to effective performance, good resolution both in the time domain as well as in the frequency domain but some drawbacks occurred like increased complexity and sensitive to direction.

### 3.3 Deep learning

Over a decade, deep learning has had a significant impact on the research quality in areas such as classification [78], detection, segmentation, and fusion [26]. A Convolutional Neural Network (CNN) is used to fuse medical images to provide efficient diagnosis [44]. In [20] Multi-Scale Local Extrema Scheme (MSLES) is used to breakdown the source pictures, making them more efficient and effective for practical implementation. To build the weight map, features are retrieved using a Siamese neural network, which results in increased Standard Deviation (SD) and Average Gradient (AG). In 2021 [43], a deep learning-based medical image fusion system that delivers unambiguous findings, edge preservation, and contrast information was developed. The study's results show a modest reduction in SD, which encourages the use of deep learning algorithms in fusion. As CNN extracts the depth and effective information to increase fusion performance, it provides a significant advantage.

**Table 2** Review on transform domain fusion techniques

| Fusion technique | Modality | Application | Advantage | Challenge |
|---|---|---|---|---|
| DCT | CT, PET and MRI [63] | Brain Image Fusion | Efficient as it decomposes the images into a series of waveforms, reducing the complexity. | Leads to undesirable side-effect including blurring. |
| DWT | MRI and PET [16], PET and MRI [58] | Brain Image Fusion | Provides good resolution both in time domain and frequency domain. Simple and easy to understand and to implement. | Lack of Shift Invariance and Poor Directional Selectivity [10, 70] |
| DTCWT | MRI and CT, MRI-T1and MRI-T2 [24] | Brain Image Fusion | Increase the directionality through the complex extension of the DWT. Enhances the fused image quality [36]. | Sensitive to sensor noise and lacking in the sense of direction [81]. |
| NSST | CT, MRI and PET [38] | Brain Image Fusion | High directional sensitivity and shift invariance at low computational complexity [27]. | Blurring effect in fused images, loss of clarity, and a reduction in contrast [74]. |
| NSCT | MRIT1-MRIT2, MRIT2 - CT [42] MRI,T1--weighted MR (MR--T1)CT, MR- GAD and PET [28] CT–MR and MR-T1–MR-T2 [14, 15] CT and MRI [41, 45, 49, 85] CT and PET | Cerebral toxoplasmo--sis, fatal stroke [42] | provided more information and have higher contrast | Computing time is high [14, 15] |

### 3.4 Hybrid fusion

The image was fused using DWT using Haar filter coefficient, which minimised spectral distortion and produced a high-quality fused image with a better Signal to Noise Ratio (SNR), but it had reduced spatial resolution [68]. SWT and curvelet transform based fusion were employed to fuse infrared and visible images [88]. PCA is used to fuse the low frequency elements, and the DTCWT is combined with it. The DTCWT is used to decompose the input images, and PCA is used instead of simple average algorithms to precisely fuse the transform domain information more precise [2]. Different input images are taken like PET and MRI, ultrasound and MRI, CT and MRI. Similarly, Shearlet Transform (ST) and PCA are integrated for fusion process. The CT and MRI images are first transformed using the ST and to the transformed images PCA was applied [64]. Then the fused image was obtained using Inverse ST and the results are compared with the DWT and CT. Two fusion techniques were combined namely DTCWT and Redundancy Discrete Wavelet Transform (RCWT) and introduced a new technique called non subsampled rotated complex wavelet transform (NSRCxWT) [16]. Average and maximum selection fusion rules are used to combine the derived spectral domain features. With the least distortion, the NSRCxWT-fused images display the best of the important structural information from CT and MRI with least distortion.

PCA and NSST were used to overcome the PCA based fusion. Images are first converted into multiscale space using NSST. Second, each subband's critical components and weights are calculated and averaged for the fusion phase. Finally, a fused image is generated by putting source images together based on weights [1]. SWT and PCA were combined for fusing images; SWT was used to decompose images, PCA was used for fusion rule to fuse sub-bands, and the fused sub-bands were reconstructed using Inverse Stationary Wavelet Transform (ISWT), which increases image detail using super-resolution method and generates smooth images [10]. The general representation of hybrid domain is shown in Fig. 10.

As the input source mask image, the IHS-Wavelet transform is processed with a discrete ripplet type-1 transform at a coefficient value of 87,000 [21]. Though the transform efficiently depicts edge information and improves the image globally at this value, it increases the computational time. The ripplet type 1 transform's output is passed via a rolling guidance filter. To fuse CT-MRI and MRI images, PCA fusion based on DWT was proposed [79], which integrates the advantage of WT into PCA fusion. In the image, the terms HF and LF stand for low and high frequencies, respectively. NSST is used here in conjunction with PCA and Particle Swarm Optimization (PSO) to find the best combination of PCA values for the fusion process [76]. NSST is used to convert perfectly recorded MR images into spectral

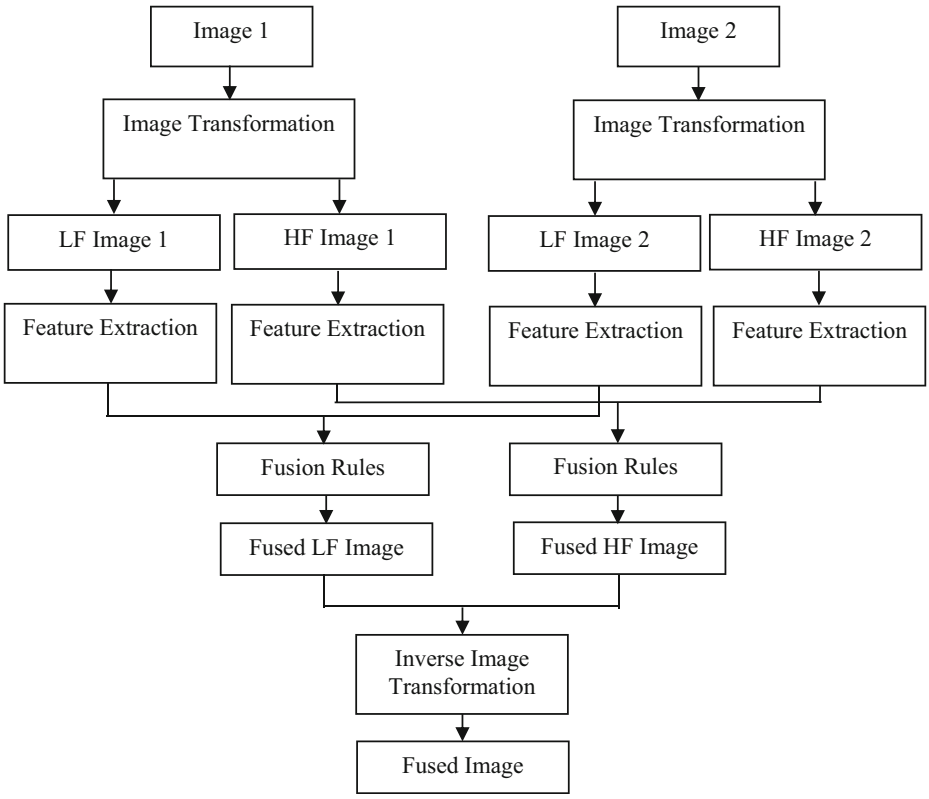


**Fig. 10** General flow diagram of hybrid domain

domain. [74], A new multimodal medical image fusion paradigm based on discrete Tchebichef moments and PCNN was implemented. To define image form, the Tchebichef moments were calculated, and the energy of blocks was shown as the sum of squared non-DC moment values. The energy of Tchebichef moments for blocks was then used to inspire the PCNN with adaptive linking power to retain the edges and textures. A fusion algorithm based on CNN in DWT domain was used [81]. Fusion between CT and MRI image, CNN in shearlet domain was proposed [31]. Table 3 shows the different hybrid fusion techniques with its applications in medical field.

In recent years deep network plays a major role for fusion process, deep network consists of input layer, convolutional layers, pooling layers, a fully connected layer, and an output layer. Deep network was used for fusing images to have improved result [44, 45, 61, 82]. Ensemble of three deep networks was proposed, in which the three deep networks are used together for fusing multi focus images [4]. Also pixel-wise CNN was introduced with two max-pooling layers and two fully connection layers and trained the model using stochastic gradient descent [75]. Then the pixel-wise convolution neural network is converted into image CNN so that the time consumption gets reduced. Deep Support Value Convolutional Neural Network (DSVCNN) was introduced to overcome the drawbacks of CNN i.e. The parameters of each filter in each layer are determined by minimising empirical risk [45]. However, it is difficult to ensure that the qualified network will have good promotional results using empirical risk minimization, since max-pooling and subsampling minimise the function map's resolution, resulting in information loss [45]. Fuzzy Transform (FTR) is capable of retaining image characteristics such as edges and change invariance. FTR was used to combine images from MRI and CT, CT and PET, MRI and SPECT, and MRI and PET [48]. To receive high-resolution images for CT and MR images, a new coupled dictionary and multi-norm constraint fusion method has been developed [80]. Fuzzy Whale Fusion (FWFusion), a hybrid approach incorporating fuzzy concepts and a novel P-Whale model for the fusion of MRI multimodal images, was introduced [57]. The P-Whale method combines the Whale Optimization Algorithm (WOA) and PSO, For successful selection of whale fusion parameters, a new multimodal medical image fusion

**Table 3** Various hybrid fusion techniques with its applications

| Fusion technique | Modalities | Applications | Observations from the survey |
|---|---|---|---|
| PCA and DWT [79] | CT and MRI | Cardiac Image Fusion | Image information such as tissues is improved in its clearly but the ventricle borders are not completely achieved |
| NSCT with DTCWT [14] | MRI and PET | Brain Image Fusion | Helps for accurate diagnosis and better treatment |
| IHS and RIM | MRI and PET | Medical Image Fusion | Good spectrum and better spatial feature detected in fused image. Resampling not required |
| DTCWT and PCA [2] | PET and MRI, CT and MRI | Brain Image Fusion | Better performing fusion for cervical spine |
| NSST with PCA [1] | MRI slice images | Diagnosis of Alzheimers disease analysis | Simple and effective fusion was achieved |
| SWT with PCA [10] | MRI and CT | Health systems | Increased information in fusion with the details of source images, less distortion and clear edges. |

algorithm based on NSST and neuro fuzzy was proposed to fuse CT and MRI images [41].

## 4 Quality metrices

Mostly the reference images are not available to evaluate the fusion algorithm. The following metrics are used to evaluate image fusion quality when a reference image isn't available. The quality metrices such as Standard deviation, Entropy, Spatial frequency, mutual information, Cross entropy, etc. are considered for the quality measure.

### 4.1 Standard deviation (SD)

It's used to work out how much contrast the fused image has. The contrast of the fused image is good and the image is clear if the standard deviation value is high [30, 33, 35, 36]. The mathematical representation of the standard deviation is referred and developed. It is given in Eq. 1.

$$S.D = \frac{\sqrt{\sum_{i=1}^{M}\sum_{j=1}^{N}(f(i,j)-\mu)^2}}{MN} \tag{1}$$

M Length of an image
N Width of an image

### 4.2 Entropy (E)

The information in the image is described by this metric. It takes the value between 0 and 8 [30, 33]. The mathematical representation of the entropy is referred and developed. It is given in Eq. 2.

$$E = -\sum_{i=0}^{L-1} P_i * \log 2\ \mathrm{P_i} \tag{2}$$

L Number of gray levels

### 4.3 Spatial frequency (SF)

It gives sharpness in fused image. Image resolution will be high if the spatial frequency of an image is large [11, 30, 36]. The mathematical representation of the spatial frequency is referred and developed. It is given in Eq. 3.

$$SF = \sqrt{RF^2 + CF^2} \tag{3}$$

RF row frequency of the image
CF column frequency of the image

### 4.4 Mutual information (MI)

Mutual Information is the metric used to measure the similarity of an image based on its pixel information. High mutual information means the better fusion which helps to extract more information [30, 33]. The mathematical representation of the mutual information is referred and developed. It is given in Eq. 4.

$$MI = I(x,f) + I(y,f) \tag{4}$$

I (x,f) and I (y,f) Conditional entropies

### 4.5 Gradient based index ($Q_{AB/F}$)

This statistic measures how well the edges of source pictures are preserved in the merged image. It ranges from 0 to 1, where higher value shows better performance. The mathematical representation of QAB/F is given in Eq. (5).

$$\begin{gathered} Q_{AB/F} = \sum_{i=1}^{M}\sum_{j=1}^{N}\left(Q_{I_1I_F}(i,j)W_{I_1}(i,j)+ \right. \\ \left. Q_{I_2I_F}(i,j)W_{I_2}(i,j)\right)/ \\ \sum_{i=1}^{M}\sum_{j=1}^{N}(W_{I_1}(i,j)+W_{I_2}(i,j)) \end{gathered} \tag{5}$$

Where Q represent the gradient information with respect to source images I1 and I2, If represent the fused image and W is a sliding window of fixed size.

## 5 Performance evaluation

The availability of fusion algorithms are large in number. The major methods are considered for performance evaluation to provide valuable information for future research. The comparisons are made in Table 4. The table represent the resultant metrics values of MI, SF, SD, and $Q_{AB/F}$. Smaller value of these metrics represent the lower performance. The higher the value indicates the better fusion perfroamce.

Fusion in shearlet domain gives higher mutual information and standard deviation when compared with the remaining six methods. It is also observed that the hybrid method like combining two different methods together have given improved values. Fuzzy in NSCT domain gives higher gradient-based index. The registrations are the major concern for image

**Table 4** Comparison of medical image fusion

| S. no. | Fusion method | MI | SF | SD | SSIM | QAB/F |
|---|---|---|---|---|---|---|
| 1. | Deep learning [32] | 4.46 | 23.07 | 74.30 | 0.74 | 0.59 |
| 2. | shearlet transform and S-PCNN [38] | - | 34.63 | 86.17 | - | - |
| 3. | NSST-PCCN [27] | 2.95 | 33.09 | 85.90 | - | 0.54 |
| 4. | Shearlet domain [29] | 4.76 | 6.46 | 96.16 | - | 0.59 |
| 5. | Fuzzy in NSCT domain [85] | 4.61 | - | 69.19 | - | 0.64 |
| 6. | Wavelet transform [83] | 2.96 | - | 3.46 | - | 0.59 |
| 7. | Multi-norm constraint fusion[80] | 2.6 | 25.10 | 68.20 | 0.69 | 0.56 |

fusion as it is the first step in image fusion. Deep learning techniques are greatly used in image fusion which can be done in different domain as future research. Data augmentation is the major concern in deep learning-based algorithms.

## 6 Conclusion

In this paper the image fusion techniques to fuse images from various medical imaging modalities have been discussed. Image fusion algorithms are very useful in the fast-growing medical imaging technology. It is complicated to develop an effective image fusion algorithm in medical sector and also to develop the devices for real-time medical imaging applications such as Brain tumor detection, etc. Since several challenges remain in development, the image fusion being proved to be a solution in medical imaging to solve such a critical problem related to a greater number of data or images. The medical image fusion techniques are increasing their trust gradually along with the growth in medical imaging modalities. The evolution of medical image fusion categorized to spatial domain, transform domain and hybrid domain. This indicates that there is a high demand for computer aided diagnosis. Though there are different methods have been proposed by different researchers, each one has its own advantages and disadvantages. In earlier days spatial domain were used in which the overall contrast of the image can be enhanced and it is very simple and less complex. Since spatial domain does not provide adequate robustness, transform domain techniques were came into picture which eases enhance edges but the spatial resolution of the fused image was less. Later on, hybrid fusion techniques were employed which combines the advantage of both spatial and transform domain (i.e., both the spatial resolution and the spectral resolution was improved). Since 2017 deep learning techniques are emerged in medical image fusion. In recent years deep network, U-Net and other deep learning techniques are used. Two different modalities or the same modality images are being fused in medical field. Mostly the MRI and CT images are fused together as MRI provides better soft tissue definition and higher spatial resolution, whereas CT provides three-dimension information with short scan time and high imaging resolutions. In near future, the fusion techniques are expected to grow rapidly in line with the innovation and advancement for the betterment of health care sector. The transform domain can be combined with machine learning technique to extract features for fusion rules.

### Declarations

**Conflict of interest** The authors certify that there is NO conflict of interest in relation to this work.

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